

Implement Client Script & UI Policy

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Implement Client Script & UI Policy (Incident)

Problem Statement

Incident records often require consistent and accurate data entry to ensure effective triage, routing, and resolution. However, relying solely on user awareness and manual checks can lead to incomplete, inconsistent, or incorrect data being submitted. This can impact reporting accuracy, SLA compliance, and overall service quality. To address this challenge, there is a need to enforce conditional field behavior and validation directly at the user interface level. Implementing standardized UI Policies and Client Scripts helps control field visibility, mandatory status, and save-time validation, ensuring that Incident records meet defined business rules before they are submitted.

Objective

The objective of the Implement Client Script & UI Policy (Incident) project is to demonstrate how ServiceNow client-side controls can be used to enforce data integrity on Incident records. This implementation showcases how UI Policies and Client Scripts can dynamically make fields mandatory, auto-populate values, control field behavior, and prevent record submission when required conditions are not met. The solution ensures that Incidents are created with complete and valid information, improving data quality and consistency across the platform.

Skills

Incident Management
UI Policy
UI Policy Actions
Client Scripts
Form Validation

Task 1: Create UI Policy on Incident

Activity 1: Create UI Policy

1. Log in to the **ServiceNow instance** with administrative or configuration access.

2. Navigate to:
System UI → UI Policies
3. Click **New** to create a new UI Policy record.
4. Fill in the following details:
 - **Name:** High Impact Control
This name clearly represents the purpose of the policy, which is to apply additional controls when an Incident has a high impact.
 - **Table:** Incident
Specifies that the UI Policy applies to Incident records.
 - **Active:** true
Enables the UI Policy so it is enforced on the Incident form.
5. Configure the **Conditions** section as follows:
 - **Field:** Impact
 - **Operator:** is
 - **Value:** 1 – High
6. This condition ensures the UI Policy is triggered only when the Incident impact is set to High.
7. And Ui policy action Assignment group
8. Check mandatory
9. Set **Reverse if false** to **true**.
This ensures that when the condition is not met, any field changes applied by the policy are automatically reverted.
10. Review the configuration to ensure accuracy.
11. Click **Submit** to save the UI Policy.

The screenshot shows the ServiceNow interface for configuring a UI Policy Action. The left sidebar contains a navigation menu with categories like 'System Mobile', 'Mobile UI', 'Service Catalog', 'System Diagnostics', 'System UI', 'MID Server', and 'System Web Services'. The main content area is titled 'UI Policy Action - assignment_group'. It features a table with columns for 'UI policy' (set to 'High Impact Control') and 'Table' (set to 'Incident [incident]'). A red box highlights the 'Field name' dropdown, which is set to 'Assignment group'. To the right, there are three dropdown menus: 'Mandatory' (set to 'True'), 'Visible' (set to 'Leave alone'), and 'Read only' (set to 'False'). Each of these three dropdowns is also highlighted with a red box. Below these settings, there are 'Update' and 'Delete' buttons.

Task 2: Create UI Policy Action – Urgency

Activity 1: Configure the Urgency Field

1. Open the UI Policy Actions

Go to the **High Impact Control** UI Policy you created earlier. Scroll down to the **UI Policy Actions** related list. This is where you define the specific behavior of fields when the policy conditions are met.

2. Create a New Action

Click **New** to create a UI Policy Action. This lets you configure how individual fields behave when the UI Policy is triggered.

3. Fill in the Action Details

- **Field name:** Urgency
Select the **Urgency** field on the Incident form. This is the field you want to control.
- **Read-only:** true
Check this option to make the Urgency field read-only whenever the UI Policy condition is met (Impact = High). Users will be able to see the field but cannot change its value.

- **Visible:** Leave this field as is (do not change).
By leaving it unchanged, the field will remain visible on the form, but its behavior will be controlled by the Read-only setting.

4. Submit the Action

Click **Submit** to save the UI Policy Action. The action is now linked to the High Impact Control UI Policy and will automatically apply to the Urgency field when an Incident has High Impact.

5. Optional Verification

To test the action:

- Open an Incident record.
- Set the **Impact** field to **High**.
- Check that the **Urgency** field becomes read-only.
- Change Impact to a different value to confirm that the field reverts back if needed.

The screenshot shows the ServiceNow interface for configuring a UI Policy Action. The left sidebar displays the navigation menu with 'UI pol' selected. The main content area is titled 'UI Policy Action - urgency'. Below the title, there is a description: 'UI policy actions specify exactly what actions to take on a specified field. The conditions specified in the UI policy determine when these actions are triggered. [More Info](#)'. The configuration form includes the following fields:

- UI policy:** High Impact Control
- Table:** Incident [incident]
- * Field name:** Urgency (highlighted with a red box)
- Application:** Global
- Mandatory:** Leave alone
- Visible:** Leave alone
- Read only:** True (highlighted with a red box)
- Clear the field value:** ☐

At the bottom of the form, there are 'Update' and 'Delete' buttons.

Task 3: Create onChange Client Script

Activity 1: Create Client Script

1. **Navigate to Client Scripts:**

Go to System UI → Client Scripts.

2. **Click New** to create a new Client Script.

3. **Fill in the details:**

- **Name:** Auto set urgency for high impact – clearly shows the purpose.
- **Table:** Incident – the script runs only on Incident records.
- **Type:** onChange – executes when a field value changes.
- **Field name:** Impact – triggers the script when Impact is updated.
- **Active:** true – ensures the script runs immediately.

4. **Paste the script** in the script editor section.

5. **Click Submit** to save the Client Script.

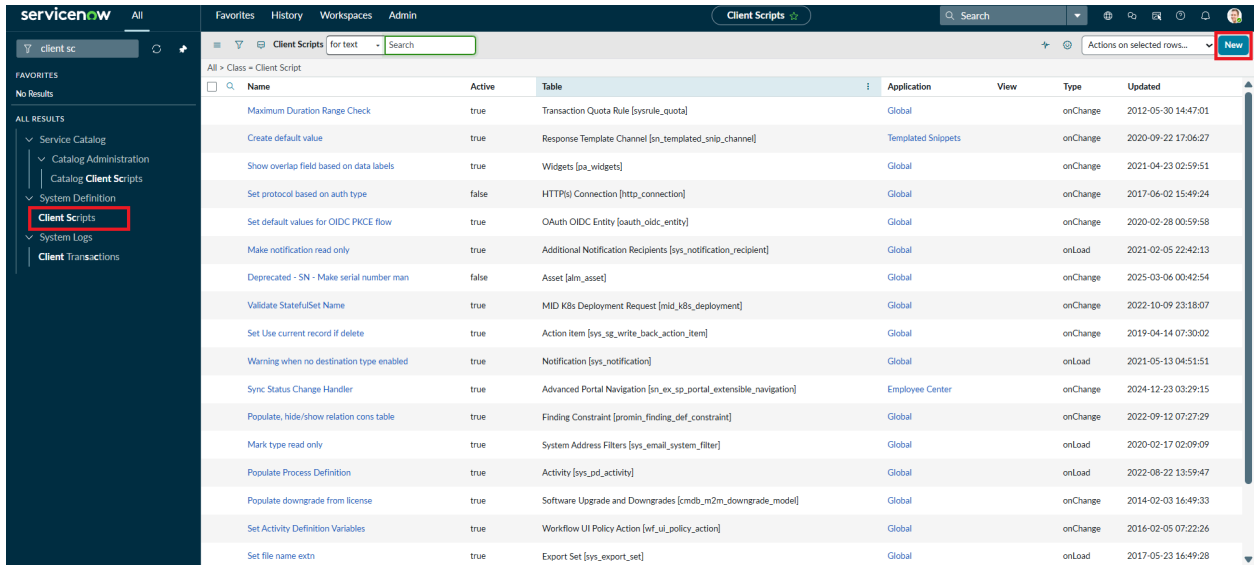
6. **Verify:** Open an Incident record and change the Impact field to **High**. The Urgency field should update automatically

Script:

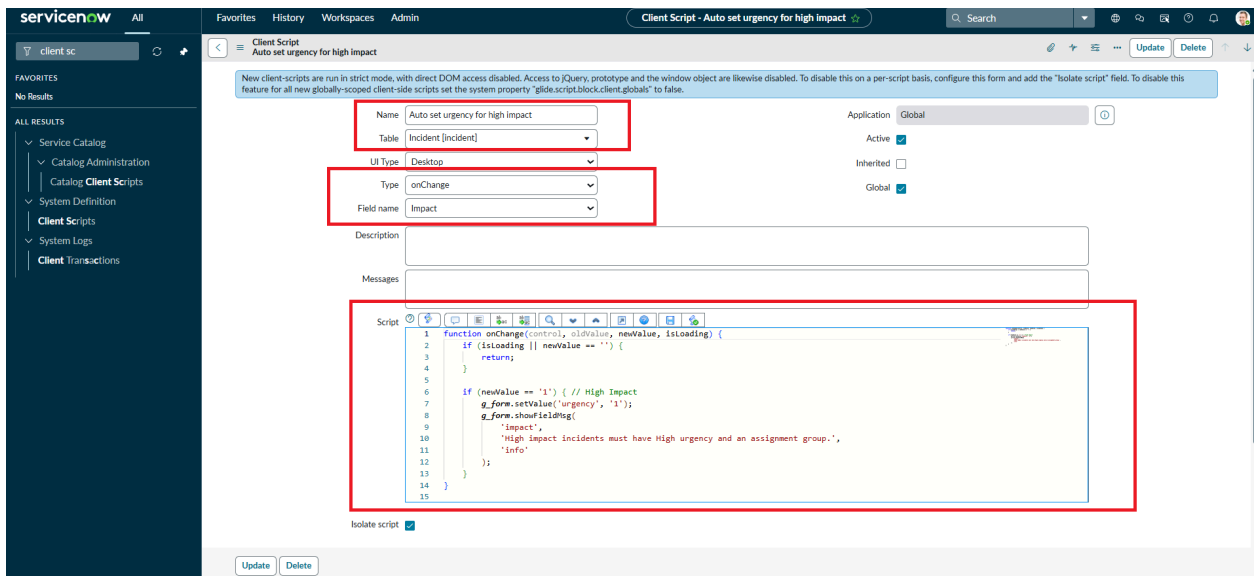
```
function onChange(control, oldValue, newValue, isLoading) {  
  
    if (isLoading || newValue === '') {  
        return;  
    }  
  
    if (newValue === '1') {  
        g_form.setValue('urgency', '1');  
        g_form.addInfoMessage('Urgency set to High for High impact incident.');    }  
}
```

}

5. Click Submit.



Name	Active	Table	Application	View	Type	Updated
Maximum Duration Range Check	true	Transaction Quota Rule [sysrule_quota]	Global		onChange	2012-05-30 14:47:01
Create default value	true	Response Template Channel [sn_templated_snip_channel]	Templated Snippets		onChange	2020-09-22 17:06:27
Show overlap field based on data labels	true	Widgets [pa_widgets]	Global		onChange	2021-04-23 02:59:51
Set protocol based on auth type	false	HTTP(s) Connection [http_connection]	Global		onChange	2017-06-02 15:49:24
Set default values for OIDC PKCE flow	true	OAuth OIDC Entity [oauth_oidc_entity]	Global		onChange	2020-02-28 00:59:58
Make notification read only	true	Additional Notification Recipients [sys_notification_recipient]	Global		onLoad	2021-02-05 22:42:13
Deprecated - SN - Make serial number man	false	Asset [alm_asset]	Global		onChange	2025-03-06 00:42:54
Validate StatefulSet Name	true	MID K8s Deployment Request [mid_k8s_deployment]	Global		onChange	2022-10-09 23:18:07
Set Use current record if delete	true	Action Item [sys_sg_write_back_action_item]	Global		onChange	2019-04-14 07:30:02
Warning when no destination type enabled	true	Notification [sys_notification]	Global		onLoad	2021-05-13 04:51:51
Sync Status Change Handler	true	Advanced Portal Navigation [sn_ex_sp_portal_extensible_navigation]	Employee Center		onChange	2024-12-23 03:29:15
Populate, hide/show relation cons table	true	Finding Constraint [promin_finding_def_constraint]	Global		onChange	2022-09-12 07:27:29
Mark type read only	true	System Address Filters [sys_email_system_filter]	Global		onLoad	2020-02-17 02:09:09
Populate Process Definition	true	Activity [sys_pd_activity]	Global		onLoad	2022-08-22 13:59:47
Populate downgrade from license	true	Software Upgrade and Downgrades [cmdb_m2m_downgrade_model]	Global		onChange	2014-02-03 16:49:33
Set Activity Definition Variables	true	Workflow UI Policy Action [wf_ui_policy_action]	Global		onChange	2016-02-05 07:22:26
Set file name extn	true	Export Set [sys_export_set]	Global		onLoad	2017-05-23 16:49:28



New client-scripts are run in strict mode, with direct DOM access disabled. Access to jQuery, prototype and the window object are likewise disabled. To disable this on a per-script basis, configure this form and add the "Isolate script" field. To disable this feature for all new globally-scoped client-side scripts set the system property "glide.script.block.client.globals" to false.

Name: Auto set urgency for high impact
Table: Incident [incident]
UI Type: Desktop
Type: onChange
Field name: Impact

Application: Global
Active: ☒
Inherited: ☐
Global: ☒

Description:
Messages:

Script

```
1 function onChange(control, oldValue, newValue, isLoading) {  
2   if (isLoading || newValue == '') {  
3     return;  
4   }  
5  
6   if (newValue == '1') { // High Impact  
7     g_form.setValue('urgency', '1');  
8     g_form.showFieldMsg(  
9       'Impact',  
10      'High Impact Incidents must have High urgency and an assignment group.',  
11      'info'  
12    );  
13  }  
14 }  
15
```

Isolate script: ☒

Update Delete

Task 4: Create onSubmit Client Script (Save Validation)

Activity 1: Create Client Script

1. **Navigate to Client Scripts:**

Go to System UI → Client Scripts in your ServiceNow instance.

2. **Click New** to create a new Client Script.

3. **Fill in the details:**

- **Name:** Prevent save if Assigned To missing – clearly indicates that the script prevents saving if the field is empty.
- **Table:** Incident – applies the script only to Incident records.
- **Type:** onSubmit – executes when the user tries to save the record.
- **Active:** true – ensures the script is enforced immediately.

4. **Paste the script** into the script editor section. This script should check if the **Assigned To** field is empty and prevent the record from saving if it is.

5. **Click Submit** to save the Client Script.

6. **Verify:** Try saving an Incident without filling the **Assigned To** field. The script should prevent the save and display a warning.

7. Paste the script:

```
function onSubmit() {  
  
    if (g_form.getValue('impact') == '1' &&  
        g_form.getValue('assigned_to') == "") {  
  
        g_form.showErrorBox(  
  
            'assigned_to',  
  
            'Assigned To is mandatory for High impact incidents.'  
  
        );  
    }  
}
```

```

    return false;
}

return true;
}

```

8. Click **Submit**.

The screenshot shows the ServiceNow interface for configuring a Client Script. The left sidebar shows the navigation menu with 'Client Scripts' selected. The main area displays the configuration for a Client Script named 'Prevent save if assigned to missing'. The configuration includes a table of 'Incident' and a type of 'onSubmit'. The script code is as follows:

```

1 function onSubmit() {
2   if (g_form.getValue('impact') == '1' &&
3       g_form.getValue('assigned_to') == '') {
4     g_form.showErrorBox(
5       'assigned_to',
6       'Assigned To is mandatory for High impact incidents.'
7     );
8     return false;
9   }
10  return true;
11 }
12
13
14

```

The 'Isolate script' checkbox is checked. The 'Update' and 'Delete' buttons are at the bottom.

Task 5: Create onCellEdit Client Script

Activity 1: Create Client Script

- Navigate to Client Scripts:**
Go to System UI → Client Scripts in your ServiceNow instance.
- Click New** to create a new Client Script.
- Fill in the details:**

- **Name:** Prevent state change via list edit – clearly indicates that the script prevents users from changing the State field directly in a list.
 - **Table:** Incident – applies the script only to Incident records.
 - **Type:** onCellEdit – runs whenever a field is edited directly in a list view.
 - **Field name:** State – the script will monitor changes to the State field.
 - **Active:** true – ensures the script is enforced immediately.
4. **Paste the script** into the script editor section. The script should prevent unauthorized changes to the State field when editing from the list view.
 5. **Click Submit** to save the Client Script.
 6. **Verify:** Open an Incident list and try to change the State field directly in the list. The script should block the change as intended.

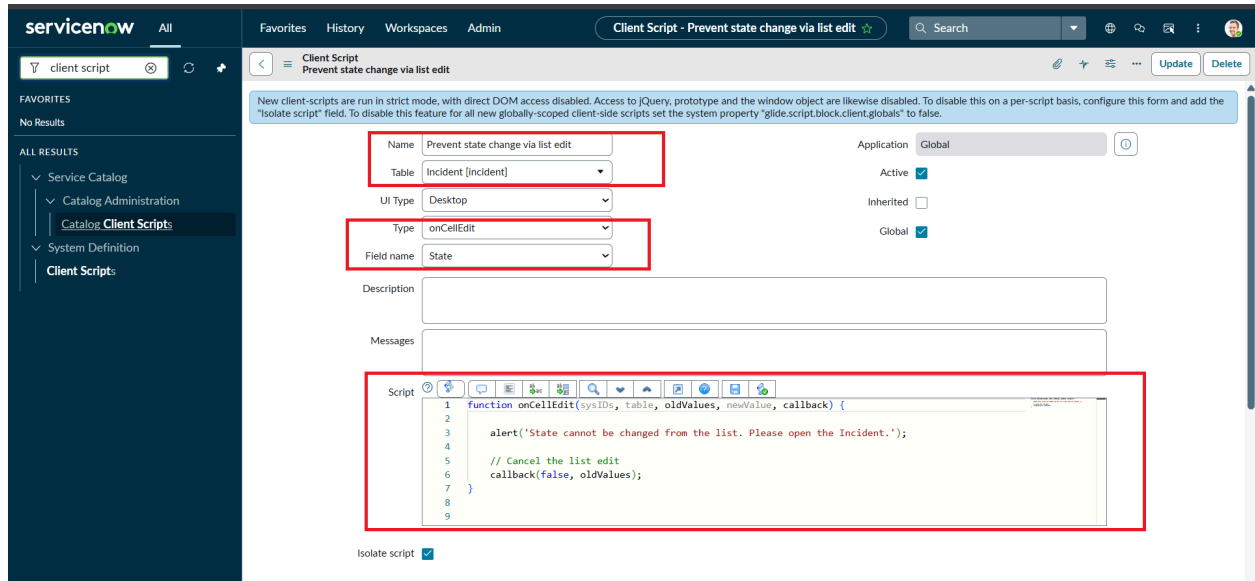
7. In the Script field, paste the following code:

```
function onCellEdit(sysIDs, table, oldValues, newValue, callback) {  
  
    alert('State cannot be updated using list editing. Please open the Incident.');
```

callback(false);

```
}
```

8. Click **Submit**.

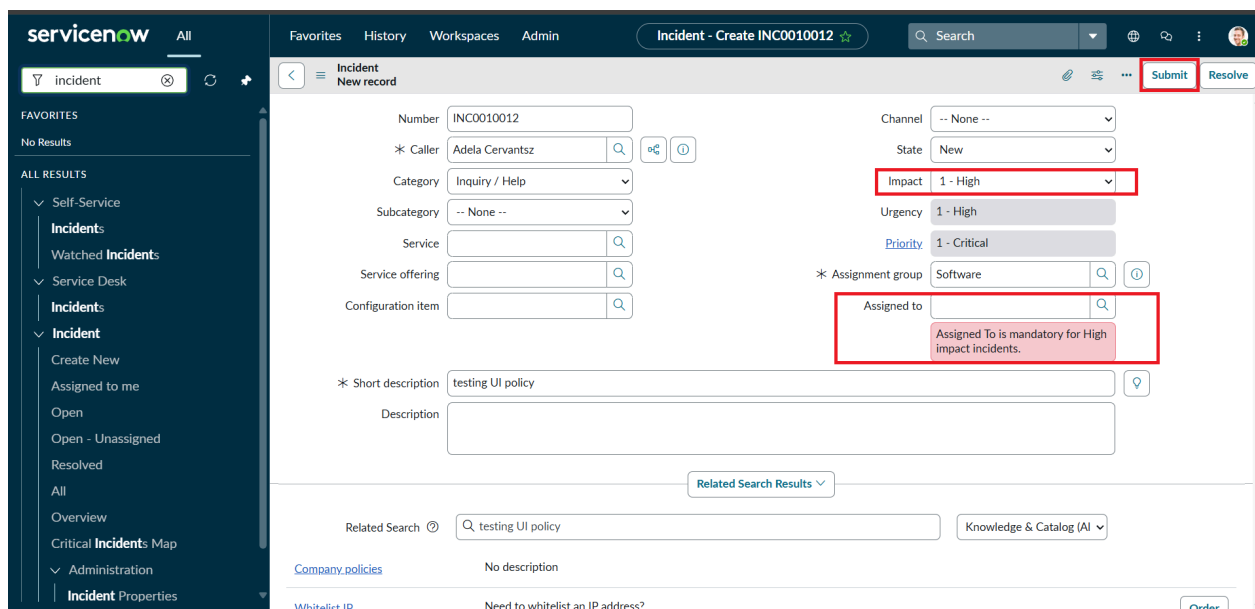
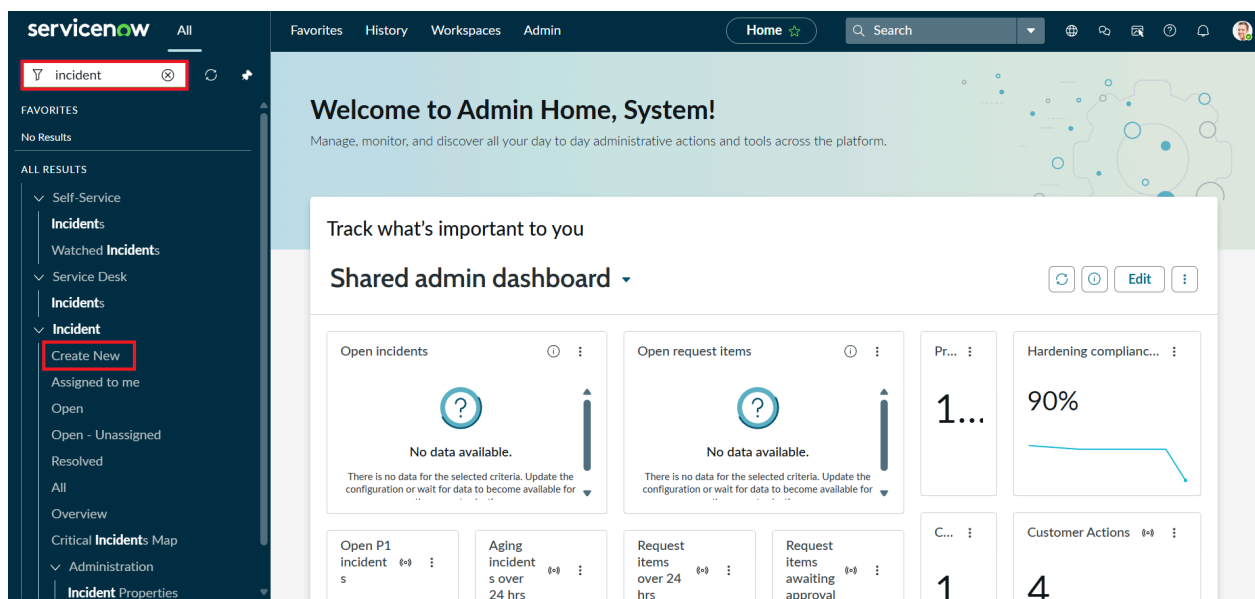


Task 6: Testing the Configuration

Activity 1: Test Mandatory Enforcement

- 1. Navigate to Incident:**
Go to Incident → Create New to open a new Incident form.
- 2. Set the Impact Field:**
Select **High** for the **Impact** field. This will trigger the UI Policy and Client Script behaviors you configured earlier.
- 3. Leave Assigned To Empty:**
Do not fill in the **Assigned To** field. This will test the onSubmit Client Script that prevents saving if Assigned To is missing.
- 4. Click Submit:**
Attempt to save the Incident record.
- 5. Verify the Result:**

- The Incident should **not be saved** because Assigned To is required when Impact is High.
- An **error message** should appear, indicating that the Assigned To field cannot be empty.
- Confirm that all UI Policy actions, such as read-only fields or automated urgency updates, are working as expected.



Activity 2: Test Successful Save

1. Fill the Assigned To Field:

Open the Incident form and select a user in the **Assigned To** field. This ensures the onSubmit Client Script condition is satisfied.

2. Click Submit:

Attempt to save the Incident record.

3. Verify the Result:

- The Incident should be **saved successfully** without any errors.
- Confirm that all other configured behaviors, such as:
 - **Urgency auto-setting** (from the onChange Client Script)
 - **Read-only fields** (from the UI Policy Action)
 - **State restrictions** (from the onCellEdit Client Script) are functioning as expected

The screenshot displays the ServiceNow 'Incident - Create INC0010012' form. The left sidebar shows the navigation menu with 'Incident' selected. The main form area contains various fields: Number (INC0010012), Caller (Adela Cervantsz), Category (Inquiry / Help), Subcategory (-- None --), Service, Service offering, Configuration item, and Short description (testing UI policy). On the right, there are dropdowns for Channel (-- None --), State (New), Impact (1 - High), Urgency (1 - High), and Priority (1 - Critical). The 'Assigned to' field is populated with 'Beth Anglin' and is highlighted with a red box. Below the form, the 'Related Search Results' section shows a search for 'testing UI policy' with results like 'Company policies', 'Whitelist IP', and 'Firewall Rule Change'.

Activity 3: Reverse Condition Test

1. Open an Existing Incident:

Go to Incident → Open and select an Incident where **Impact** is currently set to **High**.

2. Change Impact:

Update the **Impact** field from **High** to **Medium**. This will test the **Reverse if false** functionality of your UI Policy and the behavior of your Client Scripts.

3. Verify the Field Behavior:

- **Assigned To** should no longer be mandatory.
- **Urgency** should now be editable, meaning the read-only UI Policy action is reversed.

4. Save the Record:

Click **Submit** to save the changes. The record should save successfully without errors, confirming that the reverse condition works as intended.

The screenshot displays the ServiceNow Incident form for record INC0010012. The left sidebar shows the navigation menu with 'Incident' selected. The main form area contains the following fields and values:

- Number: INC0010012
- Channel: -- None --
- * Caller: Adela Cervantsz
- State: New
- Category: Inquiry / Help
- Impact: 2 - Medium (highlighted with a red box)
- Subcategory: -- None --
- Urgency: 1 - High (highlighted with a red box)
- Service:
- Priority: 2 - High
- Service offering:
- Assignment group: Software
- Configuration item:
- Assigned to: (highlighted with a red box)
- * Short description: testing UI policy
- Description:

At the bottom of the form, there is a 'Related Search Results' button and a 'Notes' tab. The 'Notes' tab is currently selected, showing a list of related records and resolution information.

Activity 4: Test List Edit Blocking

1. Navigate to the Incident List:

Go to Incident → All to view all Incident records in a list view.

2. Attempt a List Edit:

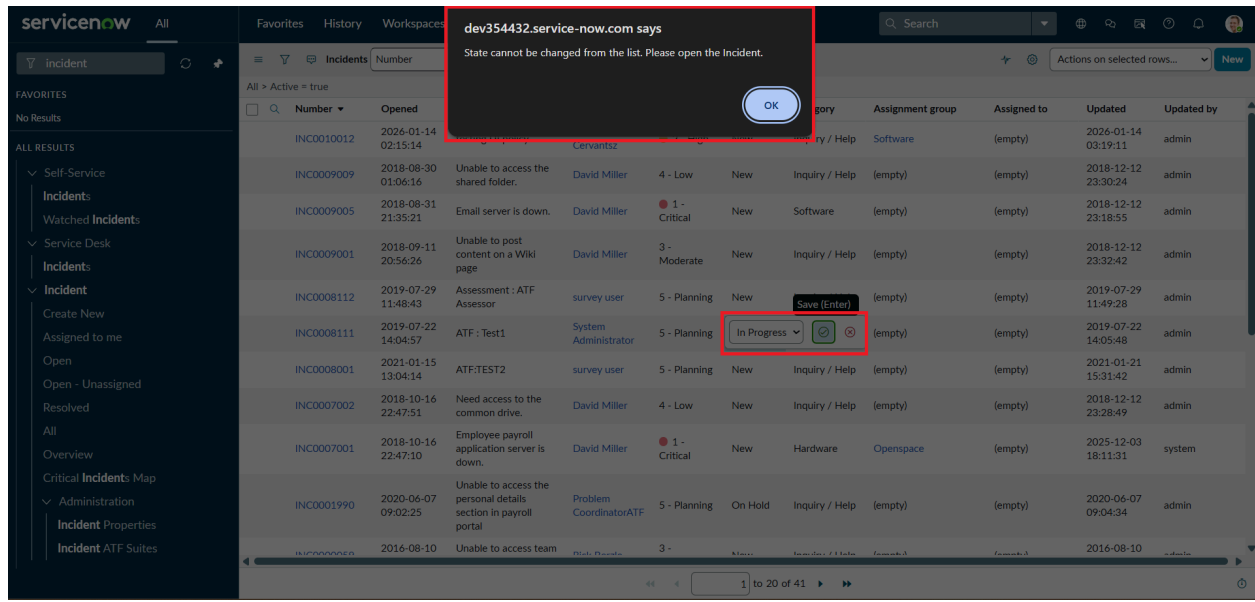
- Locate any Incident in the list.
- **Double-click the State field** to try editing it directly from the list.

3. Verify the Behavior:

- An **alert message** should appear, indicating that direct changes to the State field are not allowed.
- The **State value should remain unchanged**, confirming that the onCellEdit Client Script is working correctly.

The screenshot displays the ServiceNow Incident List interface. The left sidebar shows the navigation menu with 'Incident' selected. The main table lists incidents with columns: Number, Opened, Short description, Caller, Priority, State, Category, Assignment group, Assigned to, Updated, and Updated by. The 'State' column for incident INC0008111 is highlighted with a red box, showing the value 'New'. The bottom of the table shows pagination: '1 to 20 of 41'.

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010012	2026-01-14 02:15:14	testing UI policy	Adela Cervantsz	2 - High	New	Inquiry / Help	Software	(empty)	2026-01-14 03:19:11	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	ATF : Test1	System Administrator	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-22 14:05:48	admin
INC0008001	2021-01-15 13:04:14	ATF:TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Openspace	(empty)	2025-12-03 18:11:31	system
INC0001990	2020-06-07 09:02:25	Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07 09:04:34	admin
INC0001990	2016-08-10	Unable to access team	Problem Coordinator	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2016-08-10	admin



Activity 5: Test Form-Based Update

1. Open the Incident Record:

Navigate to the same Incident record you used in previous tests.

2. Change the State:

On the **Incident form**, update the **State** field to a new value. This tests that changes via the form (not the list) are allowed.

3. Click Update:

Save the record by clicking **Update**.

4. Verify the Result:

- The **State** change should be saved successfully.
- Confirm that other fields, such as **Assigned To** and **Urgency**, still follow the configured UI Policy and Client Script rules.

ServiceNow Incident form for INC0008111. The form includes fields for Number, Caller, Category, Subcategory, Service, Service offering, Configuration item, Short description, Description, Country, and States. A dropdown menu is open for the 'State' field, showing options like 'Add to Visual Task Board', 'Copy Incident', 'Create Incident Task', 'Create Outage', 'Create Problem', 'Create Request', 'Create Child Incident', 'Create Normal Change', 'Create Standard Change', 'Create Emergency Change', 'Refresh Impacted Services', 'Metrics Timeline', 'Follow on Live Feed', 'Analyze Access', 'Configure', 'Export', 'View', 'Create Favorite', 'Copy URL', 'Copy sys_id', 'Show XML', 'History', and 'Reload form'. The 'State' field is highlighted with a red box, and the 'In Progress' option is selected.

Number	Opened	Short description	Caller	Priority	State	Category	Assignment group	Assigned to	Updated	Updated by
INC0010012	2026-01-14 02:15:14	testing UI policy	Adela Cervantsz	2 - High	New	Inquiry / Help	Software	(empty)	2026-01-14 03:19:11	admin
INC0009009	2018-08-30 01:06:16	Unable to access the shared folder.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:30:24	admin
INC0009005	2018-08-31 21:35:21	Email server is down.	David Miller	1 - Critical	New	Software	(empty)	(empty)	2018-12-12 23:18:55	admin
INC0009001	2018-09-11 20:56:26	Unable to post content on a Wiki page	David Miller	3 - Moderate	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:32:42	admin
INC0008112	2019-07-29 11:48:43	Assessment : ATF Assessor	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2019-07-29 11:49:28	admin
INC0008111	2019-07-22 14:04:57	ATF: Test1	System Administrator	5 - Planning	In Progress	Inquiry / Help	(empty)	(empty)	2026-01-14 03:31:59	admin
INC0008001	2021-01-15 13:04:14	ATF:TEST2	survey user	5 - Planning	New	Inquiry / Help	(empty)	(empty)	2021-01-21 15:31:42	admin
INC0007002	2018-10-16 22:47:51	Need access to the common drive.	David Miller	4 - Low	New	Inquiry / Help	(empty)	(empty)	2018-12-12 23:28:49	admin
INC0007001	2018-10-16 22:47:10	Employee payroll application server is down.	David Miller	1 - Critical	New	Hardware	Opspace	(empty)	2025-12-03 18:11:31	system
INC0001990	2020-06-07 09:02:25	Unable to access the personal details section in payroll portal	Problem CoordinatorATF	5 - Planning	On Hold	Inquiry / Help	(empty)	(empty)	2020-06-07 09:04:34	admin
INC0000000	2016-08-10	Unable to access team	Philipp Pirovato	3 -	New	Inquiry / Help	Teamwork 3	Teamwork 3	2016-08-10	admin

Conclusion

The Implement Client Script & UI Policy (Incident) micro project effectively demonstrates how **UI Policies** and **Client Scripts** can work together to enforce dynamic field behavior, automate updates, and prevent incorrect data submission on Incident forms. By using onChange, onSubmit, and onCellEdit scripts alongside UI Policy actions, the project ensures fields like **Assigned To** and **Urgency** respond appropriately to changes in **Impact**, maintaining clean and

consistent data. This solution not only enforces mandatory fields and controls user interactions but also remains lightweight, efficient, and easy to implement, making it ideal for short-duration micro projects while showcasing practical best practices for improving form usability and data integrity in ServiceNow.